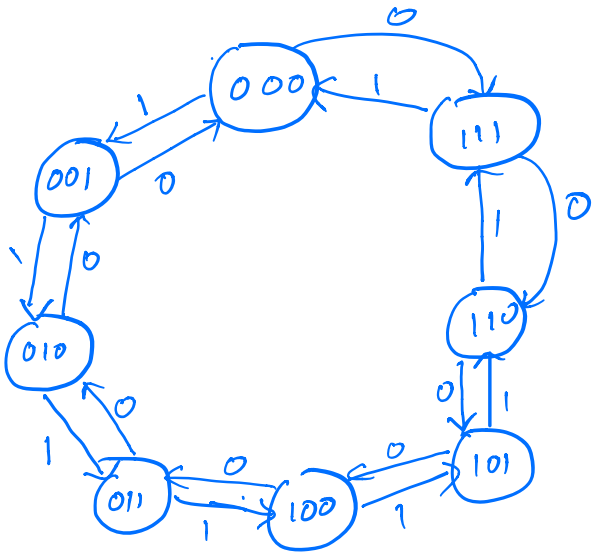
3 bit Synchronous UP-DOWN Counter:

Counter counts $0 \to 7$, $0 \to 7$ when $U = 1$, and counts $7 \to 0$, $7 \to 0$ when $U = 0$ or $D = 1$.



Pres State			Inputs		Next state			FF I/ps.		
A_2	A_1	A_0	U	$D=\bar{U}$	A_2	A_1	A_0	T_{A_2}	T_{A_1}	T_{A_0}
0	0	0	1	0	0	0	1	0	0	1
0	0	0	0	1	1	1	1	1	1	1
0	0	1	1	0	0	1	0	0	1	1
0	0	1	0	1	0	0	0	0	0	1
0	1	0	1	0	0	1	1	0	0	1
0	1	0	0	1	0	0	1	0	1	1
0	1	1	1	0	1	0	0	1	1	1
0	1	1	0	1	0	1	0	0	0	1

A_2	A_1	A_0	U	D/$\bar{U}$	A_2	A_1	A_0	T_{A_2}	T_{A_1}	T_{A_0}
1	0	0	1	0	1	0	1	0	0	1
1	0	0	0	1	0	1	1	1	1	1
1	0	1	1	0	1	1	0	0	1	1
1	0	1	0	1	1	0	0	0	0	1
1	1	0	1	0	1	1	1	0	0	1
1	1	0	0	1	1	0	1	0	1	1
1	1	1	1	0	0	0	0	1	1	1
1	1	1	0	1	1	1	0	0	0	1

Solve for T_{A_2}, T_{A_1}, T_{A_0}. Since $U=0$ is same as $D=1$ one can solve it using a 4 var K-Map in terms of $T_{A_2} = f(A_2, A_1, A_0, U)$, $T_{A_1} = g(A_2, A_1, A_0, U)$...

However we can solve separately for $U=0$, $U=1$ and combine making use of the simplification already for the UP counter as

$U=1$, $T_{A_0} = 1$, $T_{A_1} = A_0$, $T_{A_2} = A_1 A_0$

When $U=0$ / $D=1$, taking only the prime rows of the above state transition table we get.

$U=0$, $T_{A_0} = 1$, $T_{A_1} = \bar{A_0}$, $T_{A_2} = \bar{A_0}\bar{A_1}$ from K-Map.

Combine $U=1$, $U=0$ to get:

$$T_{A_0} = U(1) + \bar{U}(1) = U + \bar{U}$$

$$T_{A_1} = U A_0 + \bar{U}\bar{A}_0 = U \odot A_0$$

$$T_{A_2} = U(A_1 A_0) + \bar{U}(\bar{A}_0 \bar{A}_1)$$

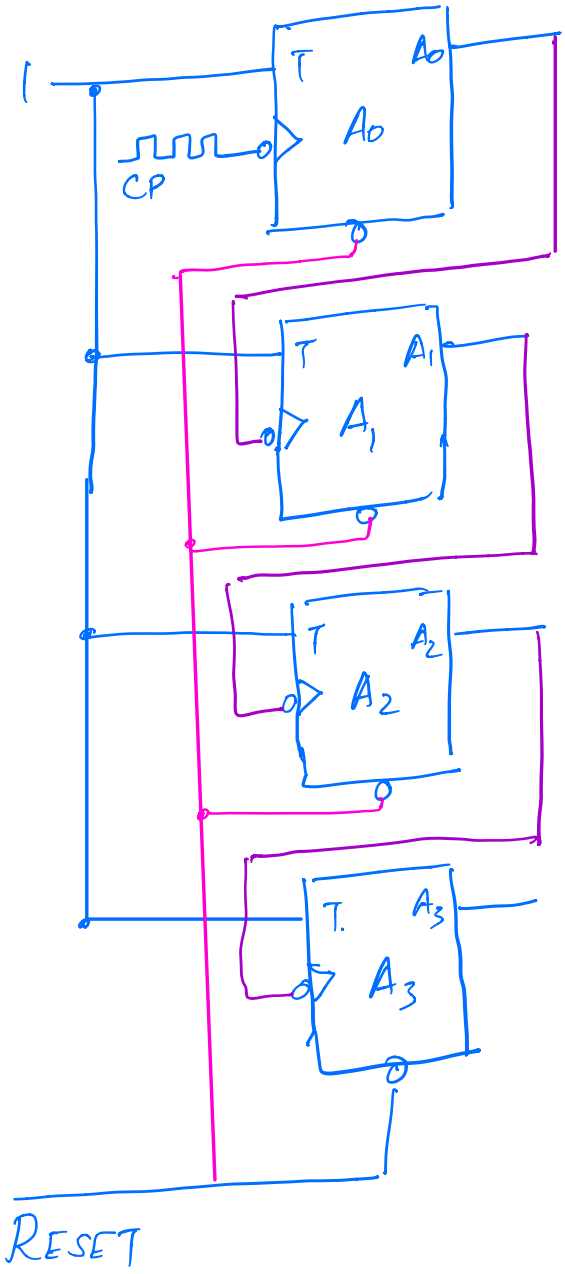
This is analogous to Figure 6.13 except that this is for a 3 bit U/D, while Fig 6.13 is for a 4 bit U/D

ASYNCHRONOUS COUNTERS:

→ Clock is not synchronous or the SAME for all the flip flops. Typically the clock is 1/2, 1/4 of the original base clock to the FF inputs.

Ripple Counter: Consists of a series of complementary FFs with the o/p of each FF is connected to the clk i/p of the next higher FF.

Falling Edge Triggered 4 bit Ripple Counter.



A_0 toggles at the falling edge of the CP.

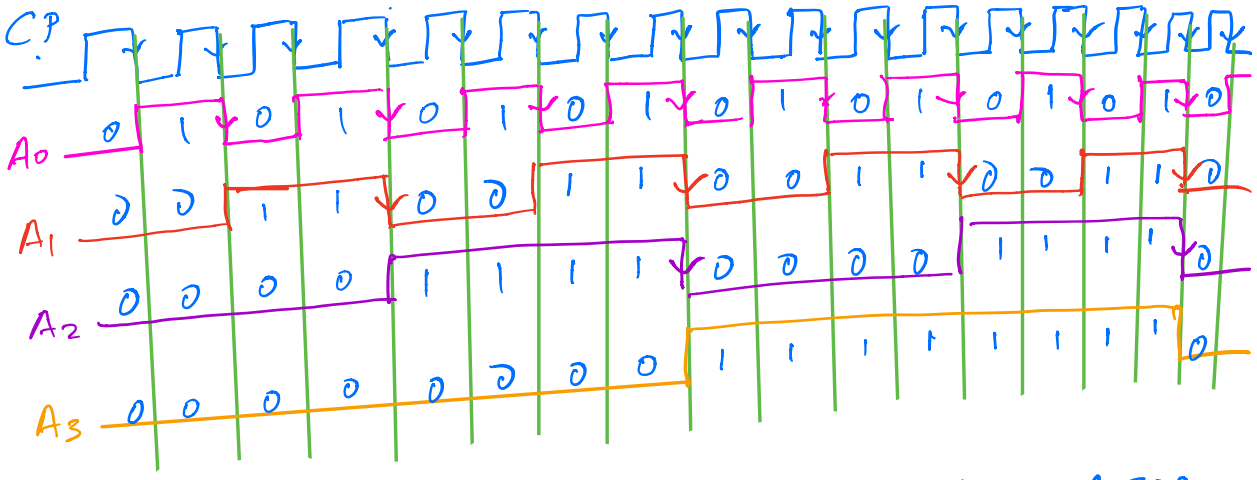
A_1 toggles when?

A_2 toggles ??

A_3 toggles ??

So, what is the clock frequency of A_1, A_2, A_3 with respect to the base clock of A_0?

$= 1/2\,(CP),\ 1/4\,(CP),\ \frac{1}{8}(CP)$



Why did we NOT use rising edge triggered FF?

BCD Ripple Counter.